Case Study 1: Student Result Management System

Problem

A college wants to develop a Java program to calculate the result of a student. The student has marks in five subjects.

The program should:

1. Accept student name and marks.
2. Calculate total and percentage.
3. Display grade according to:
    - Percentage ≥ 90 → A+
    - Percentage ≥ 80 → A
    - Percentage ≥ 70 → B
    - Percentage ≥ 60 → C
    - Percentage ≥ 50 → D
    - Below 50 → F
4. Display whether the student has passed or failed.

Case Study 2: Bank Account

Problem

A bank wants to create a simple account management program.

Create a class BankAccount containing:

- Account number
- Account holder name
- Balance

Implement methods:

- deposit()
- withdraw()
- displayBalance()

The withdrawal should not be allowed if the requested amount is greater than the available balance.

Case Study 3: Employee Salary Management

Problem

An organization maintains employee information.

Create an Employee class with:

- Employee ID
- Name
- Basic salary

Calculate:

- HRA = 20% of basic salary
- DA = 10% of basic salary
- Gross salary = Basic + HRA + DA

Display the complete salary details.

Case Study 4: Online Shopping Cart

Problem

An online shopping application stores products in a shopping cart.

Create a Product class containing:

- Product ID
- Product name
- Price
- Quantity

Calculate the total cost of each product and the total bill.

If the total bill exceeds ₹5,000, provide a **10% discount**.

Case Study 5: Library Management System

Problem

A college library wants to maintain books.

Create a Book class containing:

- Book ID
- Book title
- Author

- Availability status

Implement methods:

- issueBook()
- returnBook()
- displayBook()

A book cannot be issued if it is already issued.

Case Study 6: Hospital Patient Management

Problem

A hospital maintains patient information.

Create a Patient class with:

- Patient ID
- Name
- Age
- Temperature

The program should determine whether the patient has fever.

If temperature is greater than **100.4°F**, display "Fever"; otherwise display "Normal".

Case Study 7: ATM Simulation

Problem

Design an ATM program that:

1. Checks PIN.
2. Allows withdrawal.
3. Allows deposit.
4. Displays balance.
5. Allows maximum 3 incorrect PIN attempts.

Case Study 8: Employee Inheritance

Problem

A company has two types of employees:

- Employee

- Manager

Every employee has a name and salary. A manager additionally receives a bonus.

Use **inheritance** to implement the system.

Case Study 9: University Course Registration

Problem

A university allows students to register for courses.

A student can register for a maximum of **5 courses**.

Create a class Student with methods:

- registerCourse()
- displayCourses()

If the student tries to register for more than 5 courses, display an appropriate message.

Case Study 10: E-Commerce Payment System — Polymorphism

Problem

An e-commerce website supports different payment methods:

- Credit Card
- UPI
- Net Banking

Create a common Payment interface with a pay() method.

Implement the interface using different classes.